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--- Summary ---

Province of the
EASTERN CAPE
EDUCATION

NASIONALE
SENIOR SERTIFIKAAT

GRAAD 11

NOVEMBER 2012

WISKUNDIGE GELETTERDHEID V2
MEMORANDUM

PUNTE : 100

Simbool	Verduideliking
M	Metode
MA	Metode met akkuraatheid
CA	Voortgesette akkuraatheid
A	Akkuraatheid (Antwoord)
C	Omskakeling
S	Vereenvoudiging
RT/RG /RM	Lees van tabel/Lees van grafiek/Lees van kaart
F	Kies van korrekte formule
SF	Substitusie in formule
J/O	Mening
P	Penalisering vir geen eenhede, verkeerde afronding, ens.

■
(5) 1:A
(radius)
1:SF
1:A

1:C (cm3
na l)

1:J

1.4
LU1
AS
11.1.1 1.4.1 500 g = ½ kg
½ x R 67 = R 33,50 ■ (1) 1:A

LU1
AS
11.1.2 1.4.2 BTW bedrag = (R33,50 + R12,59) ■ x 0,14
= R 46,09 x 0,14 ■
= R 6,45 ■
(3) 1:M

(korrekte
waardes)
1:M
(x 14%)
1:A

LU1
AS
11.1.3 1.4.3 Alhoewel nartjies vrugte is, is dit in hierdie geval
geprosesseer (geblik), terwyl die lemoene ■ vars
produk is .
■ (2) 1:A
1:R

LU4
AS
11.4.5 3.1.2 10 ■■
(2) 2:A

LU4
AS
11.4.5 3.1.3 4 ■■
(2) 2:A

LU4
AS
11.4.5 3.1.4 10 ■■
(2) 2:A

LU4
AS
11.4.5 3.1.5 ¼ ■■ OF 0,25 ■■ OF 25% ■■
(2) 1:A (teller)
1:A (noemer)

LU4
AS
11.4.5 3.1.6 1/16 ■■ OF 0,063 ■■ OF 6,3% ■■
(2) 1:A (teller)
1:A (noemer)

3.2
LU2
AS
11.2.1 3.2.1 (a) s = 5t + 2 c + 3p ■■■
(3) 3:F

LU2
AS
11.2.1 (b) s = 5t + 2 c + 3p
= (5 x 6) + (2 x 5) + (3 x 3) ■
= 30 + 10 + 9
= 49 ■ (2) 1:SF
(korrekte
waardes)
1:CA

LU2
AS
11.2.1 3.2.2 Vir 1 strafskop ■
(1) 1:A

3.3
LU1

AS
11.1.1 3.3.1 1 ZAR (Suid -Afrikaanse Rand) = 0,15761 NZD
Kategor ie B = 123 NZD ■

ZAR = 123 NZD/0,15761 NZD ■
= 780,4073346
= 780,41 ■ (3) 1:RT (123)

1:M

1:A

LU1
AS
11.1.1 3.3.2 1 NZD = R6,3450 ZAR
200 NZD
ZAR = 200 x 6 ,3450 ■
= R1 269 ■ (2) 1:M
1A
[23]

(NOVEMBER 2012) WISKUNDIGE GELETTERDHEID V2 7

VRAAG 4

4.1
LU2
AS
11.2.3 4.1.1 Kosprys vir 1 CD = (80 – 30)/10 ■OF Kosprys vir 1 CD = (130 – 30)/20 ■
= 50/10
= 100/20
= R5 ■ = R5 ■

Kosprys vir 1 CD = (180 – 30)/30 ■OF Kosprys vir 1 CD = (230 – 30)/10 ■
= 150/30
= 200/40
= R5 ■ = R5 ■

Kosprys vir 1 CD = (280 – 30)/50■OF Kosprys vir 1 CD = (330 – 30)/60 ■
= 250/50
= 300/60
= R5 ■ = R5 ■

Kosprys vir 1 CD = (380 – 30)/70■OF Kosprys vir 1 CD = (430 – 30)/80 ■
= 350/70
= 400/80
= R5 ■ = R5 ■

Kosprys vir 1CD = (480 – 30)/90 ■
= 450/90
= R 5 ■ (2) 1:M
1:A

LU2
AS
11.2.3 4.1.2 Verkoopprys van 1 CD = 60/10■ OF Verkoopprys van = 120/20 ■
= R6 ■
= R6 ■

Verkoopprys van 1 CD = 180/30■ OF Verkoopprys van 1 CD = 240/40
■
= R6 ■
= R6 ■

Verkoopprys van 1 CD = 300/50 ■ OF Verkoopprys van 1 CD = 360/60 ■
= R6 ■ = R6 ■

Verkoopprys van 1 CD = 420/70 ■ OF Verkoopprys van 1CD = 480/80 ■
= R 6 ■ = R 6 ■
OF
Verkoopprys van 1CD = 540/90 ■
= R6 ■ (2) 1:M
1:A

LU2
AS
11.2.3 4.1.3 % Wins = (Inkomste – Uitgawes) x 100
Uitgawes
= 6 – 5 x 100
5
= 1/5 x 100 ■
= 20% ■ (2) 1:M
1:A

LU2
AS
11.2.1 4.1.4 30 ■ ; 180 ■
(2) 1:A
(30)
1:A
(180)

8 WISKUNDIGE GELETTERDHEID V2 (NOVEMBER 2012)

LU2
AS
11.2.1 4.1.5 Gelykbreekpunt ■
Vir 30 CD's is die inkomste en uitgawes presies dieselfde
(R180) .
van binnesirkel
= r2 – r2
= 3,14 x 5,9 cm x 5,9 cm – 3,14 x 0,75 cm x 0,75 cm ■
= 109,3034 cm2 – 1,76625 cm2 ■
= 107,53715 cm2
= 107,54 cm2 ■ (5) 1:C
(mm
na cm)
1:A
(vind r)

1:SF
1:S
1:CA

[25]

TOTA AL: 100

Province of the
EASTERN CAPE
EDUCATION

NATIONAL

SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2012

MATHEMATICAL LITERACY P2
MEMORANDUM

MARKS: 100

Symbol	Explanation
M	Method
MA	Method with accuracy
CA	Consistent accuracy
A	Accuracy
C	Conversion
S	Simplification
RT/RG /RM	Reading from a table/Reading from a graph /Read from map
F	Choosing the correct formula
SF	Substitution in a formula
J	Justification
P	Penalty, e.g.

■ (2) 1:A
1:R

2.3
LO4
AS
11.4.5

P(silver) = 3/14 ■■ OR 0,214 ■■ OR 21,4% ■■
(2) 1:A
(numerator)
1:A
(denominator)
[21]

QUESTION 3

3.1
LO4
AS
11.4.5 3.1.1 5 ■
Teams cannot play against themselves ■ (2) 1:A
1:R

LO4
AS
11.4.5 3.1.2 10 ■■
(2) 2:A

LO4
AS
11.4.5 3.1.3 4 ■■
(2) 2:A

LO4
AS
11.4.5 3.1.4 10 ■■
(2) 2:A

LO4
AS
11.4.5 3.1.5 ¼ ■■ OR 0,25 ■■ OR 25% ■■
(2) 1:A
(numerator)
1:A
(denominator)

LO4
AS
11.4.5 3.1.6 1/16 ■■ OR 0,063 ■■ OR 6,3% ■■
(2) 1:A
(numerator)
1:A
(denominator)

3.2
LO2
AS
11.2.1 3.2.1 (a) s = 5t + 2 c + 3p ■■■
(3) 3:F

LO2
AS
11.2.1 (b) s = 5t + 2 c + 3p
= (5 x 6) + (2 x 5) + (3 x 3) ■
= 30 + 10 + 9
= 49 ■ (2) 1:SF (correct
values)
1:CA

LO2
AS
11.2.1 3.2.2 For 1 penalty ■
(1) 1:A

3.3
LO1
AS
11.1.1 3.3.1 1 ZAR (South African Rand) = 0,15761 NZD
Category B = 123 NZD ■

ZAR = 123 NZD/0,15761 NZD ■
= 780,4073346
= 780,41 ■ (3) 1:RT (123)

1:M

1:A

LO1
AS
11.1.1 3.3.2 1 NZD = R6,3450 ZAR
200 NZD
ZAR = 200 x 6 ,3450 ■
= R1 269 ■ (2) 1:M
1A
[23]

QUESTION 4

4.1

LO2
AS
11.2.3 4.1.1 Cost price for 1 CD = (80 – 30)/10 ■OR Cost price for 1 CD = (130 – 30)/20 ■
= 50/10
= 100/20
= R5 ■ = R5 ■

Cost price for 1 CD = (180 – 30)/30 ■OR Cost price for 1 CD = (230 – 30)/10 ■
= 150/30
= 200/40
= R5 ■ = R5 ■

Cost price for 1 CD = (280 – 30)/50■OR Cost price for 1 CD = (330 – 30)/60 ■
= 250/50
= 300/60
= R5 ■ = R5 ■

Cost price for 1 CD = (380 – 30)/70■OR Cost price for 1 CD = (430 – 30)/80 ■
= 350/70
= 400/80
= R5 ■ = R5 ■

Cost price for 1 CD = (480 – 30)/90 ■
= 450/90
= R5 ■ (2) 1:M
1:A

4.1.2 Selling price of 1 CD =60/10■ OR Selling price of 1 CD = 120/20 ■
= R6 ■ = R6 ■

Selling price of 1 CD = 180/30■ OR Selling price of 1 CD = 240/40 ■
= R6 ■ =
R6■

Selling price of 1 CD = 300/50 ■ OR Selling price of 1 CD = 360/60 ■
= R6 ■ = R6 ■

Selling price of 1CD = 420/70 ■ OR Selling price of 1CD = 480/80 ■
= R6 ■ = R6 ■
OR
Selling price of 1CD = 540/90 ■
= R6 ■ (2) 1:M
1:A

LO2
AS
11.2.3 4.1.3 Percentage profit = (Income – Expenses) x 100
Expenses
= 6 – 5 x 100
5
% profit = 1/5 x 100 ■
= 20% ■ (2) 1:M
1:A

LO2
AS
11.2.1 4.1.4 30 ■ ; 180 ■
(2) 1:A
(30)
1:A
(180)

LO2
AS
11.2.1 4.1.5 Break -even point ■
For 30 CD’s the income and expenses are exactly the same
(R180).
Serves 4 people

1 lb (pound) 2 oz (ounce) Ostrich Fillet
1½ tsp beef stock powder
Juice and grated peel of 1 large orange
60 m■ vinegar
1 tbsp honey
2 tbsp olive oil
1 can naartjies
Salt and freshly ground pepper to taste

1.1.1 While Grecthen is preparing the recipe, the only measuring spoon that she has is a 5 m■ teaspoon (tsp).

--- Full Combined Text ---

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GRAAD 11

NOVEMBER 2012

WISKUNDIGE GELETTERDHEID V2
MEMORANDUM

PUNTE : 100

Simbool	Verduideliking
M	Metode
MA	Metode met akkuraatheid
CA	Voortgesette akkuraatheid
A	Akkuraatheid (Antwoord)
C	Omskakeling
S	Vereenvoudiging
RT/RG /RM	Lees van tabel/Lees van grafiek/Lees van kaart
F	Kies van korrekte formule
SF	Substitusie in formule
J/O	Mening
P	Penalisering vir geen eenhede, verkeerde afronding, ens.
R	(Afronding / Rede)

Hierdie memorandum bestaan uit 8 bladsye .

2 WISKUNDIGE GELETTERDHEID V2 (NOVEMBER 2012)

VRAAG 1

1.1
LU3

AS
11.3.2 1.1.1 1 tlp = 5 m■ OF tsp = 15/5 ■
1 etlp = 15 m■ = 3 tlp x 2■
15 m■ = 3 tlp ■ = 6 tlp ■
Daarom is 30 ml = 3 tlp x 2 ■
= 6 tlp ■ (3) 1:C (m■
na tlp)
1:M (x2)
1:A

LU3
AS
11.3.2 1.1.2 1 blikkie = 4 mense
20 mense ■
4 mense
= 5 ■
Om 20 mense te bedien sal 5 blikkies nartjies benodig
word. (2) 1:M
(20/4)
1:A

LU3
AS
11.3.2 1.1.3 ºF = ºC x 1,8 + 32
= 220º x 1,8 + 32 ■
= 396º ■ + 32
= 428º ■
400º ≠ 430 º
Nee, Gretchen het die oond op die verkeerde temperatuur
ºC gestel . ■ (4) 1:SF
1:S
1:A
1:R (10º)

1.2
LU 3
AS
11.3.2 1 lb (pond) 2 oz (onse) Volstruisfilet
1 lb = 0,45359 kg
1 oz = 0,0625 lb

0,0625 lb x 0,45359 = 0,028349375 kg ■x 2 ■
= 0,05669875 kg
Kilogram volstruisfilet = 1 lb + 2 oz
= 0,45359 kg + 0,05669875 kg ■
= 0,51028875 kg ■
= 0,5 kg ■ (5) 1:C (lb na
kg)
1: M (x2)

1:M
1:A
1:R

1.3
LU3
AS
11.3.1

Volume = r2h

= 3,14 x 11 cm x 11 cm x 9 cm ■■
= 3419,46 cm3 ■

As 1 000 cm3 = 1l dan is
3419,46 cm3/1 000 = 3,4 ■■
Ja, die kastrol is groot genoeg vir die gekookte dis . ■
(5) 1:A
(radius)
1:SF
1:A

1:C (cm3
na l)

1:J

1.4
LU1
AS
11.1.1 1.4.1 500 g = ½ kg
½ x R 67 = R 33,50 ■ (1) 1:A

LU1
AS
11.1.2 1.4.2 BTW bedrag = (R33,50 + R12,59) ■ x 0,14
= R 46,09 x 0,14 ■
= R 6,45 ■
(3) 1:M
(korrekte
waardes)
1:M
(x 14%)
1:A

LU1
AS
11.1.3 1.4.3 Alhoewel nartjies vrugte is, is dit in hierdie geval
geprosesseer (geblik), terwyl die lemoene ■ vars
produk is . ■■
OF
Geen BTW is betaalbaar op vars produkte soos lemoene
nie, maar wanneer dit geblik word, is belasbaar . ■■
(Enige ander relevante verduideliking .) (2) 2:A

LU1
AS
11.1.3 1.4.4 Omdat 1 en 2 sent muntstukke nie meer in sirkulasie is
nie, word die finale bedrag afgerond tot die naaste 5
sent. ■■
(Enige ander relevante verduideliking .) (2) 2:A

LU1
AS
11.1.3 1.4.5 Geen kleingeld is aan Gretchen verskuldig nie,
omdat sy slegs die verskuldigde bedrag betaal . ■■ (2) 2:A

LU1
AS
11.1.1 1.4.6 Middag ■ 16:42 ■
(2) 2:A
[31]

VRAAG 2

2.1
LU1
AS
11.1.1 2.1.1 Waarde van deposito = R250 000 x 0,16 ■
= R40 000 ■ (2) 1:M
1:A

LU1
AS
11.1.1 2.1.2 P = R250 000 – R40 000
= R210 000 ■
n = 72 / 12
= 6 jare ■
i = 9,5 / 100
= 0,095

= 210 000 (1 + 6 x 0,095) ■
= 210 000 (1+ 0,57)
= 210 000 (1,57) ■
= R329 700 ■ (5) 1:A (P -
waarde)

1:A (n -
waarde)

1:SF

1:S
1:A

LU1
AS
11.1.1 2.1.3 I = A – P
= R329 700 – R210 000 ■
= R119 700 ■ (2) 1:M
1:CA

2.2
LU2
AS
11.2.1 2.2.1
= 250 000 (1 – 0,2)2 ■
= 250 00 (0,8)2
= 250 000 (0,64) ■
= R160 000 ■
OF

= 200 000 (1 – 0,2)1 ■
= 200 000 (0,8)1■
= R160 000 ■ (3) 1:SF
1:S
1:A

LU1
AS
11.2.2 2.2.2
1 punt
vir elk

(0,250
000)
(1,200
000)
(2,160
000)
(3,128
000)
(4,102
400)

(5)

LU1
AS
11.2.3 2.2.3 Indirekte of Omgekeerde eweredigheid ■
Soos die jare toeneem, so neem die waardes van
die motor af . ■ (2) 1:A
1:R

2.3
LU4
AS
11.4.5
P (silwer) = 3/14 ■■ OF 0,214 ■■ OF 21,4% ■■
(2) 1:A (teller)
1:A
(noemer)
[21]

VRAAG 3

3.1
LU4
AS
11.4.5 3.1.1 5 ■
Spanne kan nie teen hulself speel nie . ■ (2) 1:A
1:R

LU4
AS
11.4.5 3.1.2 10 ■■
(2) 2:A

LU4
AS
11.4.5 3.1.3 4 ■■
(2) 2:A

LU4
AS
11.4.5 3.1.4 10 ■■
(2) 2:A

LU4
AS
11.4.5 3.1.5 ¼ ■■ OF 0,25 ■■ OF 25% ■■
(2) 1:A (teller)

1:A (noemer)

LU4
AS
11.4.5 3.1.6 1/16 ■■ OF 0,063 ■■ OF 6,3% ■■
(2) 1:A (teller)
1:A (noemer)

3.2
LU2
AS
11.2.1 3.2.1 (a) s = 5t + 2 c + 3p ■■■
(3) 3:F

LU2
AS
11.2.1 (b) s = 5t + 2 c + 3p
= (5 x 6) + (2 x 5) + (3 x 3) ■
= 30 + 10 + 9
= 49 ■ (2) 1:SF
(korrekte
waardes)
1:CA

LU2
AS
11.2.1 3.2.2 Vir 1 strafskop ■
(1) 1:A

3.3
LU1
AS
11.1.1 3.3.1 1 ZAR (Suid -Afrikaanse Rand) = 0,15761 NZD
Kategor ie B = 123 NZD ■

ZAR = 123 NZD/0,15761 NZD ■
= 780,4073346
= 780,41 ■ (3) 1:RT (123)

1:M

1:A

LU1
AS
11.1.1 3.3.2 1 NZD = R6,3450 ZAR
200 NZD
ZAR = 200 x 6 ,3450 ■
= R1 269 ■ (2) 1:M
1A
[23]

VRAAG 4

4.1
LU2
AS
11.2.3 4.1.1 Kosprys vir 1 CD = (80 – 30)/10 ■OF Kosprys vir 1 CD = (130 – 30)/20 ■
= 50/10
= 100/20
= R5 ■ = R5 ■

Kosprys vir 1 CD = (180 – 30)/30 ■OF Kosprys vir 1 CD = (230 – 30)/10 ■
= 150/30
= 200/40
= R5 ■ = R5 ■

Kosprys vir 1 CD = (280 – 30)/50■OF Kosprys vir 1 CD = (330 – 30)/60 ■
= 250/50
= 300/60
= R5 ■ = R5 ■

Kosprys vir 1 CD = (380 – 30)/70■OF Kosprys vir 1 CD = (430 – 30)/80 ■
= 350/70
= 400/80
= R5 ■ = R5 ■

Kosprys vir 1CD = (480 – 30)/90 ■
= 450/90
= R 5 ■ (2) 1:M
1:A

LU2
AS
11.2.3 4.1.2 Verkoopprys van 1 CD = 60/10■ OF Verkoopprys van = 120/20 ■
= R6 ■
= R6 ■

Verkoopprys van 1 CD = 180/30■ OF Verkoopprys van 1 CD = 240/40
■
= R6 ■
= R6 ■

Verkoopprys van 1 CD = 300/50 ■ OF Verkoopprys van 1 CD = 360/60 ■
= R6 ■ = R6 ■

Verkoopprys van 1 CD = 420/70 ■ OF Verkoopprys van 1CD = 480/80 ■
= R 6 ■ = R 6 ■
OF
Verkoopprys van 1CD = 540/90 ■
= R6 ■ (2) 1:M
1:A

LU2
AS
11.2.3 4.1.3 % Wins = (Inkomste – Uitgawes) x 100
Uitgawes
= 6 – 5 x 100
5
= 1/5 x 100 ■
= 20% ■ (2) 1:M
1:A

LU2
AS
11.2.1 4.1.4 30 ■ ; 180 ■
(2) 1:A
(30)
1:A
(180)

LU2
AS
11.2.1 4.1.5 Gelykbreekpunt ■

Vir 30 CD’s is die inkomste en uitgawes presies dieselfde
(R180) . ■ (2) 1:A
1:R

LU2
AS
11.2.3 4.1.6 Voor die gelykbreekpunt is die inkomste minder as die
uitgawes . ■■
OF
Voor die gelykbreekpunt is die uitgawes meer as die
inkomste . ■■ (2) 2:A

LU2
AS
11.2.3 4.1.7 Daar is ■ aanvangskoste van R30 (vervoerkoste) . ■■
(2) 2:A

4.2
LU4
AS
11.4.3 4.2.1 25% van die verkope was 15 en minder CD’s verkoop vir die
maand . ■■ (2) 2:A

LU4
AS
11.4.3 4.2.2 75% van die verkope was 37 en meer CD’s verkoop vir die
maand . ■■ (2) 2:A

LU4
AS
11.4.3 4.2.3 Ja ■
Meeste van die CD’s wat hy verkoop het, is meer as 15
(75%) . ■ (2) 1:A
1:R

4.3
LU3
AS
11.3.1
11.3.2 Deursnit van buitesirkel = 118 mm = 11,8 cm ■
Radius van buitesirkel = 5,9 cm ■

Radius van b innesirkel = 0,75 cm

Opp. van CD = Opp . van buitesirkel – Opp. van binnesirkel
= r2 – r2
= 3,14 x 5,92 – 3,14 x 0,752 ■
= 109,30 cm2 – 1,77 cm2 ■
= 107,53 cm2 ■
OF
Opp van CD = Opp . van buitesirkel – Opp. van binnesirkel
= r2 – r2
= 3,14 x 5,9 cm x 5,9 cm – 3,14 x 0,75 cm x 0,75 cm ■
= 109,3034 cm2 – 1,76625 cm2 ■
= 107,53715 cm2
= 107,54 cm2 ■ (5) 1:C
(mm
na cm)
1:A
(vind r)

1:SF
1:S
1:CA

[25]

TOTA AL: 100

Province of the
EASTERN CAPE
EDUCATION

NATIONAL
SENIOR CERTIFICATE

GRADE 1 1

NOV EMBER 201 2

MATHEMATICAL LITERACY P2
MEMORANDUM

MARKS: 100

Symbol	Explanation
M	Method
MA	Method with accuracy
CA	Consistent accuracy
A	Accuracy
C	Conversion
S	Simplification
RT/RG /RM	Reading from a table/Reading from a graph /Read from map
F	Choosing the correct formula
SF	Substitution in a formula
J	Justification
P	Penalty, e.g. for no units, incorrect rounding off etc.
R	Rounding Off/Reason

This memorandum consists of 8 pages.

QUESTION 1

1.1
LO3
AS
11.3.2 1.1.1 1 tsp = 5 ml OR tsp = 15/5 ■
1 tbsp = 15 ml = 3 tsp x 2■
15 ml = 3 tsp ■ = 6 tsp ■
Therefore 30 ml = 3 tsp x 2 ■
= 6 tsp ■ (3) 1:C (m■ to
tsp)
1:M (x2)

1:A

LO3
AS
11.3.2 1.1.2 1 can = 4 people

20 people ■
4 people
= 5 ■
Therefore to serve 20 people, 5 cans of naartjies will be used . (2) 1:M (20/4)
1:A

LO3
AS
11.3.2 1.1.3 ºF = ºC x 1,8 + 32
= 220º x 1,8 + 32 ■
= 396º ■ + 32
= 428º ■
400º ≠ 430 º
No, Gretchen did not set the oven’s temperature in ºC
Correctly . ■ (4) 1:SF
1:S
1:A
1:R

1.2
LO3
AS
11.3.2 1 lb (pound) 2 oz (ounce) Ostrich Fillet
1 lb = 0,45359 kg
1 oz = 0,0625 lb

0,0625 lb x 0,45359 = 0,028349375 kg ■x 2 ■
= 0,05669875 kg
Kilograms of ostrich fillet = 1 lb + 2 oz
= 0,45359 kg + 0,05669875 kg ■
= 0,51028875 kg ■
= 0,5 kg ■ (5) 1:C (lb to kg)
1: M (x2)

1:M
1:A
1:R

1.3
LO3
AS
11.3.1

Volume = r2h
= 3,14 x 11 cm x 11 cm x 9 cm ■■
= 3 419,46 cm3 ■

If 1 000 cm3 = 1l then
3419,46 cm3/1 000 = 3,4 ■■
Yes, the casserole dish will be big enough to transfer the cooked meal . ■
(5) 1:A
(radius)
1:SF
1:A

1:C (cm3 to
l)

1:J

1.4
LO1
AS
11.1.1 1.4.1 500 g = ½ kg
½ x R67 = R33,50 ■ (1) 1:A

LO1
AS
11.1.2 1.4.2 VAT amount = (R33,50 + R12,59) ■ x 0,14
= R46,09 x 0,14 ■
= R6,45 ■
(3) 1:M
(correct
values)
1:M
(x 14%)
1:A

LO1
AS
11.1.3 1.4.3 Although naartjies are fruit, in this case it processed
(canned), while the oranges are fresh produce . ■■
OR
No VAT is paid on fresh produce such as the oranges,
but when it is canned VAT will be paid . ■■
(Accept any other relevant explanation .) (2) 2:A

LO1
AS
11.1.3 1.4.4 As the 1 and 2 cent coins are no more in circulation, the
final amount is rounded to the nearest 5 cent . ■■
(Accept any other relevant explanation .) (2) 2:A

LO1
AS
11.1.3 1.4.5 No change is due to Gretchen as she paid only the due
amount . ■■ (2) 2:A

LO1
AS
11.1.1 1.4.6 Afternoon ■ 16:42 ■
(2) 2:A
[31]

QUESTION 2

2.1
LO1
AS
11.1.1 2.1.1 Value of deposit = R250 000 x 0,16 ■
= R40 000 ■ (2) 1:M
1:A

LO 1
AS
11.1.1 2.1.2 P = R250 000 – R40 000
= R210 000 ■
n = 72 / 12

= 6 years ■
i = 9.5 / 100
= 0,095

= 210 000 (1 + 6 x 0,095) ■
= 210 000 (1+ 0,57)
= 210 000 (1,57) ■
= R329 700 ■ (5) 1:A (P -
value)

1:A (n -
value)

1:SF
1:S
1:A

LO1
AS
11.1.1 2.1.3 I = A – P
= R329 700 – R210 000 ■
= R119 700 ■ (2) 1:M
1:CA

2.2
LO2
AS
11.2.1 2.2.1
= 250 000 (1 – 0,2)2 ■
= 250 00 (0,8)2
= 250 000 (0,64) ■
= R160 000 ■
OR

= 200 000 (1 – 0,2)1 ■
= 200 000 (0,8)1■
= R160 000 ■ (3) 1:SF
1:S
1:A

LO1
AS
11.2.2 2.2.2
1 mark
for
each
for
(0,250
000)
(1,200
000)
(2,160
000)
(3,128
000)
(4,102
400)

(5)

LO1
AS
11.2.3 2.2.3 Indirect or Inverse proportion ■
As the years increase, the values of the car
decrease . ■ (2) 1:A
1:R

2.3
LO4
AS
11.4.5

P(silver) = 3/14 ■■ OR 0,214 ■■ OR 21,4% ■■
(2) 1:A
(numerator)
1:A
(denominator)
[21]

QUESTION 3

3.1
LO4
AS
11.4.5 3.1.1 5 ■
Teams cannot play against themselves ■ (2) 1:A
1:R

LO4
AS
11.4.5 3.1.2 10 ■■
(2) 2:A

LO4
AS
11.4.5 3.1.3 4 ■■
(2) 2:A

LO4
AS
11.4.5 3.1.4 10 ■■
(2) 2:A

LO4
AS
11.4.5 3.1.5 ¼ ■■ OR 0,25 ■■ OR 25% ■■
(2) 1:A
(numerator)
1:A
(denominator)

LO4
AS
11.4.5 3.1.6 1/16 ■■ OR 0,063 ■■ OR 6,3% ■■
(2) 1:A
(numerator)
1:A
(denominator)

3.2

LO2
AS
11.2.1 3.2.1 (a) s = 5t + 2 c + 3p ■■■
(3) 3:F

LO2
AS
11.2.1 (b) s = 5t + 2 c + 3p
= (5 x 6) + (2 x 5) + (3 x 3) ■
= 30 + 10 + 9
= 49 ■ (2) 1:SF (correct
values)
1:CA

LO2
AS
11.2.1 3.2.2 For 1 penalty ■
(1) 1:A

3.3
LO1
AS
11.1.1 3.3.1 1 ZAR (South African Rand) = 0,15761 NZD
Category B = 123 NZD ■

ZAR = 123 NZD/0,15761 NZD ■
= 780,4073346
= 780,41 ■ (3) 1:RT (123)

1:M

1:A

LO1
AS
11.1.1 3.3.2 1 NZD = R6,3450 ZAR
200 NZD
ZAR = 200 x 6 ,3450 ■
= R1 269 ■ (2) 1:M
1A
[23]

QUESTION 4

4.1
LO2
AS
11.2.3 4.1.1 Cost price for 1 CD = (80 – 30)/10 ■OR Cost price for 1 CD = (130 – 30)/20 ■
= 50/10
= 100/20
= R5 ■ = R5 ■

Cost price for 1 CD = (180 – 30)/30 ■OR Cost price for 1 CD = (230 – 30)/10 ■
= 150/30
= 200/40
= R5 ■ = R5 ■

Cost price for 1 CD = (280 – 30)/50■OR Cost price for 1 CD = (330 – 30)/60 ■
= 250/50
= 300/60
= R5 ■ = R5 ■

Cost price for 1 CD = (380 – 30)/70■OR Cost price for 1 CD = (430 – 30)/80 ■
= 350/70
= 400/80
= R5 ■ = R5 ■

Cost price for 1 CD = (480 – 30)/90 ■
= 450/90
= R5 ■ (2) 1:M
1:A

4.1.2 Selling price of 1 CD =60/10■ OR Selling price of 1 CD = 120/20 ■
= R6 ■ = R6 ■

Selling price of 1 CD = 180/30■ OR Selling price of 1 CD = 240/40 ■
= R6 ■ =
R6■

Selling price of 1 CD = 300/50 ■ OR Selling price of 1 CD = 360/60 ■
= R6 ■ = R6 ■

Selling price of 1CD = 420/70 ■ OR Selling price of 1CD = 480/80 ■
= R6 ■ = R6 ■
OR
Selling price of 1CD = 540/90 ■
= R6 ■ (2) 1:M
1:A

LO2
AS
11.2.3 4.1.3 Percentage profit = (Income – Expenses) x 100
Expenses
= 6 – 5 x 100
5
% profit = 1/5 x 100 ■
= 20% ■ (2) 1:M
1:A

LO2
AS
11.2.1 4.1.4 30 ■ ; 180 ■
(2) 1:A
(30)
1:A
(180)

LO2
AS
11.2.1 4.1.5 Break -even point ■
For 30 CD’s the income and expenses a re exactly the same
(R180). ■ (2) 1:A
1:R

LO2
AS
11.2.3 4.1.6 Before the breakeven point the income is less than the
expenses . ■■
OR
Before the breakeven point the expenses is more than the
income ■■ (2) 2:A

LO2

AS
11.2.3 4.1.7 There is an initial cost of R30 (transport cost) ■■
(2) 2:A

4.2
LO4
AS
11.4.3 4.2.1 25% of the sales were 15 and less CD’s for the month ■■
(2) 2:A

LO4
AS
11.4.3 4.2.2 75% of the sales were 37 and more CD’s for the month ■■
(2) 2:A

LO4
AS
11.4.3 4.2.3 Yes. ■
Most of the CD’s he sold is above 15 (75%) . ■ (2) 1:A
1:R

4.3
LO3
AS
11.3.1
11.3.2 Diameter of outer circle = 118 mm = 11,8 cm ■
Radius of outer circle = 5,9 cm ■

Radius of inner circle = 0,75 cm

Area of CD = Area of outer circle – Area of inner circle
= r2 – r2
= 3,14 x 5,92 – 3,14 x 0,752 ■
= 109,30 cm2 – 1,77 cm2 ■
= 107,53 cm2 ■
OR
Area of CD = Area of outer circle – Area of inner circle
= r2 – r2
= 3,14 x 5,9 cm x 5,9 cm – 3,14 x 0,75 cm x 0,75 cm ■
= 109,3034 cm2 – 1,76625 cm2 ■
= 107,53715 cm2
= 107,54 cm2 ■ (5) 1:C (mm
to cm)
1:A
(finding
r)

1:SF
1:S
1:CA

[25]

TOTAL: 100

Province of the
EASTERN CAPE
EDUCATION

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GRADE 1 1

NOV EMBER 201 2

MATHEMATICAL LITERACY P2

MARKS: 100

TIME: 2½ hours

This question paper consists of 11 pages and 1 page annexure .

MLITE2

INSTRUCTIONS AND INFORMATION

Read the following instructions carefully before answering the questions.

1. This question paper consists of FOUR questions. Answer ALL the questions.

2. QUESTIONS 2. 2.2 must be answered on the attached ANNEXURE A . Write your name in the spaces provided and hand in the annexure with the ANSWER BOOK.

3. Number the questions correctly according to the numbering system used in this question paper.

4. An approved calculator (non -programmable and non -graphical) may be used, unless stated otherwise.

5. ALL calculations must be shown clearly.

6. ALL the final answers must be rounded off to TWO decimal places, unless stated otherwise.

7. Start EACH question on a NEW page.

8. Write neatly and legibly.

QUESTION 1

Gretchen has decided to eat healthier by substituting her red meat diet with that of ostrich meat. Ostrich meat is fat -free, low in calories and cholesterol, while being rich in protein.

1.1 Gretchen is collecting all the recipes for cooking with ostrich meat. The following is a recipe for Ostrich Fillet in Naartjie Sauce .

Serves 4 people

1 lb (pound) 2 oz (ounce) Ostrich Fillet
1½ tsp beef stock powder
Juice and grated peel of 1 large orange
60 m■ vinegar
1 tbsp honey
2 tbsp olive oil

1 can naartjies
Salt and freshly ground pepper to taste

1.1.1 While Grecthen is preparing the recipe, the only measuring spoon that she has is a 5 m■ teaspoon (tsp). The recipe requires 2 tablespoons (tbsp) of olive oil. How many teaspoons of olive oil will Grecthen use if ? (3)

1.1.2 How many cans of naartjie s will be used if Gretchen prepares the recipe for 20 people? (2)

1.1.3 According to the recipe Gretchen has to preheat the oven to 400 ºF (Fahrenheit). Her stove’s temperature is in degrees Celsius (ºC) and she sets it to 220 ºC. Show by using the formula below whether Gretchen used the correct temperature setting in ºC.

Formula;

Give your answer to the nearest 10º . (4)

1.2 In South Africa meat is sold in kilograms (kg). Convert the ostrich fillet needed for the recipe to determine how many kilograms of ostrich fillet Gretchen has to buy. Give your final answer to 1 decimal place.

Use the following information:

1 lb = 0,45359 kg
1 oz = 0,0625 lb (5)

4 MATHEMATICAL LITERACY P2 (NOVEMBER 2012)

1.3 When the dish is completely cooked, it has a capacity of 1,5 litre. Gretchen wants to transfer it to a round serving dish as illustrated in the diagram below (not drawn to scale). The diameter of the serving dish is 22 cm and the height is 9 cm.

(5)

Use the following information to determine if the serving dish will be big enough for the cooked meal. Give your final answer to 1 decimal place.
where
1 000 cm3 = 1 ■

1.4 Gretchen received the following till slip for some of the items she bought for the recipe from SAVE -A-LOT Supermarket.

1.4.1 Determine why Gretchen only paid R33,50 for the ostrich fillet. (1)

1.4.2 Show how the Value Added Tax (VAT) of R 6,45 was calculated. (3)

1.4.3 The naartjies that Gretchen bought was taxed, but not the oranges. Give an explanation why the naartjies was taxed, but not the oranges . (2)

22 cm
9 cm
SAVE -A-LOT Supermarket
Thank you for shopping with us

Ostrich fillet 500 g
@ 67,00/ kg 33,50
0,275 kg Oranges
@ 5,99/ kg # 1,65
Naartjies 12,59

Due VAT incl. 52,54
Cash 52,50
Rounding 0,04
Items 3

----------------- TAX INVOICE ------------------
VAT included @ 14% 6,45
Non Taxable Items #
---------- VAT REG NO. 123456789 --------

Please retain slip as your guarantee

12/09/2011 16:42

1.4.4 Explain what the meaning of rounding R 0,04 on the till slip is. (2)

1.4.5 Explain why the till slip does not show any change due to Gretchen. (2)

1.4.6 What time of the day did Gretchen make the purchase? Give evidence from the till slip to justify your answer. (2)

[31]

QUESTION 2

2.1 Mondo is driving a second hand car for the past four years. Over these four years he managed to save 16% as a deposit of the price of the brand new car he wants to buy. He is interested in buying a 2011 Chevrolet Orlando

1,8 model.

Mondo visited the Delta Motor Dealer to purchase the car. The price of the car is R250 000.
For the purpose of this we are going to focus on the price of the car only (no other initial cost).

2.1.1 Calculate the amount Mondo saved for the deposit. (2)

2.1.2 Mondo managed to get a loan from a bank for the outstanding amount. He will repay the loan over 72 months at 9,5% per annum simple interest. Calculate how much Mondo will pay if it takes him exactly 72 months to repay the loan.

Use the formula: where;
,
,

(5)

2.1.3 How much interest will Mondo pay on the loan? (2)

2.2 When a new car is bought, the owner must be aware of the fact that the value of the car depreciates. As soon as you drive your brand new car out of the showroom, it does not have the same value you bought it for.

Mondo learnt that the value of his car depreciates at a rate of 20% per annum. The following table shows the relationship between the value of the car and the number of years.

Table 1

Year	0	1	2	3	4
Value in Rand	250 000	200 000	A	128 000	102 400

2.2.1 Calculate the missing value (A) by using the formula;
where;

(3)

2.2.2 Use the information in the table to draw a graph (ANNEXURE A) that shows the relationship between the value of the car and the number of years. (5)

2.2.3 What type of proportion is illustrated by the table or graph? Give a reason for your answer. (2)

2.3 In the showroom there were different colours of the Chevrolet Orlando that Mondo wants to buy (2 red, 3 silver, 4 grey and 5 black). What is the probability that Mondo will choose a silver Chevrolet Orlando ? (2)
[21]

QUESTION 3

3.1 During the first round of the 2011 Rugby World Cup the competing countries played in groups. They played every other team in their group only once. One of the groups (GROUP D) in the table below shows the possible games. The teams in GROUP D were Fiji (F), South Africa (SA), Samoa (S), Wales (W) and Namibia (N). Use the table to answer the questions below.

Table 2

Teams	F	SA	S	W	N
F	FF	FSA	FS	FW	FN
SA	SAF	SASA	SAS	SAW	SAN
S	SF	SSA	SS	SW	SN
W	WF	WSA	WS	WW	WN
N	NF	NSA	NS	NW	NN

3.1.1 How many of the matches were not possible? Explain your answer. (2)

3.1.2 How many matches did not take place since the teams only played each other once? (2)

3.1.3 How many matches did each team play? (2)

3.1.4 What is the total number of matches played during this group stage of the Rugby World Cup? (2)

3.1.5 What is the probability that a team won at least ONE of their matches? (2)

3.1.6 Calculate the probability that a team will win all their matches in the group stage and proceed to the quarter finals . Take into consideration that not one of the matches ended in a draw. (2)

3.2 The final score of the match between South Africa and Fiji was 49 – 3 in favour of South Africa. A try is worth 5 points, a conversion 2 points and a penalty 3 points.

3.2.1 (a) Write down a formula to show how the final score of a rugby match is calculated. Use (s) for final score, (t) for tries, (c) for conversions and (p) for penalties. (3)

(b) Use your formula in QUESTION 3.2.1 (a) to show how South Africa's final score was calculated if they scored 6 tries,
5 conversions and 3 penalties. (2)

3.2.2 How did Fiji score their points? (1)

3.3 The match between South Africa and Fiji was played at Wellington Regional Stadium in Wellington City, New Zealand. The ticket prices for this stadium for this specific match are listed in the table below in New Zealand Dollar (NZD).

Table 3

Category A	Category B	Category C	Category D
153 NZD	123 NZD	97 NZD	66 NZD

3.3.1 Trevor, a South African citizen, bought a Category B ticket via the internet. When the ticket was bought the exchange rate was
1 ZAR (South African Rand) = 0,15761 NZD.

Calculate how much he paid for this ticket in Rand (ZAR). (3)

3.3.2 When Trevor came back from New Zealand, he still had 200

NZD left and exchanges it to Rand (ZAR). The exchange rate at the time of exchanging was 1 NZD = R 6,3450 (ZAR). What was the value of 200 NZD in Rand (ZAR)? (2)
[23]

QUESTION 4

4.1 Shameeg, a Grade 11 learner, decided to sell Compact Discs (CDs). He wants to use the profit that he makes to fund his matric farewell costs. He bought CDs with R500 he received for his birthday. Take into consideration that Shameeg spends R30 towards transport costs.

The following graph shows his expenses and income for the CDs.

4.1.1 Calculate the cost of one CD. (2)

4.1.2 Calculate the selling price of one CD. (2)

4.1.3 With reference to your answers in QUESTIONS 4.1.1 and 4.1.2, calculate the percentage profit he makes on one CD.
Use the formula:
(–)
(2)

4.1.4 Give the coordinates of the point where the two graphs intersect. (2)

4.1.5 Give a name for the point where the two graphs intersect and explain what is meant by this intersection . (2)

4.1.6 Explain what can be noticed before the point that you have mentioned in QUESTION 4.1.5. Your explanation must refer to income and expenses. (2)
0100200300400500600
0 10 20 30 40 50 60 70 80 90Amount in Rand
Number of CD's Expenses and Income of CDs
Expenses
Income

4.1.7 The graph for expenses does not start at the origin (0;0). Explain why the graph does not start at the origin. (2)

4.2 Shameeg used the data of his sales for the month and drew the five - number summary below to analyse his sales of CDs during one month. Use this five-number summary to answer the questions below.

Table 4

Minimum	5
Quartile 1	15
Quartile 2	24
Quartile 3	37
Maximum	42

4.2.1 What can we deduce from the first quartile (15) on the sales of the CDs? (2)

4.2.2 What can be deduced from the third quartile (37) on the sales of the number of CDs? (2)

4.2.3 Would you say Shameeg had a good month with the sales of the number of CDs? Explain your answer. (2)

4.3 The following is a drawing of a CD (not drawn to scale).

118 mm

0,75
cm

Shameeg read an article which stated that the area of a CD is 107,53 cm2. Use the diagram above where the diameter of outer circle is 118 mm and the radius of the inner circle 0,75 cm to help Shameeg to see if the statement is correct.
Use the formula; where .
Give you final answer in cm2. (5)
[25]

TOTAL: 100
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ANNEXURE A

QUESTION 2.2.2

NAME:

050 000100 000150 000200 000250 000
0 1 2 3 4 5Value in Rand
Years Value of car over years